Laborator 10

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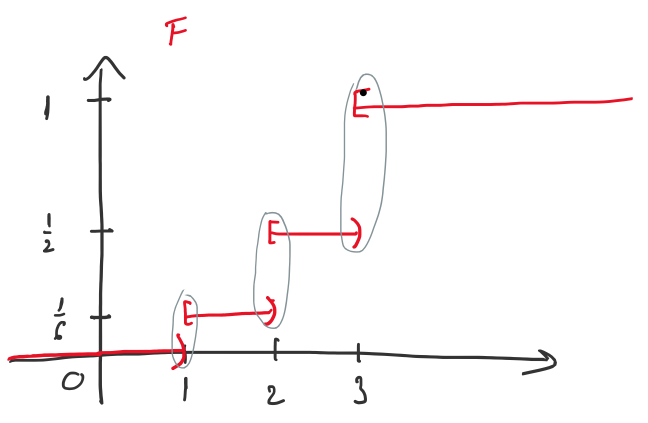
Funcții de repartiție a unei variabile aleatoare discrete

(c.d.f)

$F : \mathbb{R} \to [0,1]$ $\quad F(x) = \mathbb{P}(X \le x)$

ex: $X : \begin{pmatrix} \overset{x}{\}} 1 & \overset{x}{\}} 2 & \overset{x}{\}} 3 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}$ $\quad F(x) = \begin{cases} 0 & , x < 1 \\ \frac{1}{6} & , 1 \le x < 2 \\ \frac{1}{2} = \frac{1}{6} + \frac{1}{3} & , 2 \le x < 3 \\ 1 & , x \ge 3 \end{cases}$



$$\boxed{\mathbb{P}(a < X \le b) = F(b) - F(a)}$$

$$\begin{aligned} \mathbb{P}(X \in (a, b]) &= \mathbb{P}(X \in (-\infty, b] \setminus (-\infty, a]) \\ &= \mathbb{P}(\underbrace{X \in (-\infty, b]}_{A} \setminus \underbrace{X \in (-\infty, a]}_{B}) \\ &= \mathbb{P}(X \in (-\infty, b]) - \mathbb{P}(X \in (-\infty, a]) \\ &= F(b) - F(a) \end{aligned}$$

$$\boxed{\begin{aligned} \mathbb{P}(a \le X \le b) &= F(b) - F(a) + \mathbb{P}(X = a) \\ \mathbb{P}(a < X < b) &= F(b) - F(a) - \mathbb{P}(X = b) \\ \mathbb{P}(a \le X < b) &= F(b) - F(a) - \mathbb{P}(X = b) + \mathbb{P}(X = a) \end{aligned}}$$

V. a. condiționate

X \ Y	y_1	y_2	y_j	y_m	P
x_1	π_{11}	π_{12}	π_{1j}	π_{1m}	p_1
x_2	π_{21}	π_{22}	π_{2j}	π_{2m}	p_2
x_i	π_{i1}	π_{i2}		π_{im}	p_i
x_n	π_{n1}	π_{n2}	π_{nj}	π_{nm}	p_n
q	q_1	q_2	q_j	q_m	

$$X \mid Y = y_j : \begin{pmatrix} x_1 & x_2 & \dots & x_n \\ \frac{\pi_{1j}}{q_j} & \frac{\pi_{2j}}{q_j} & \dots & \frac{x_{nj}}{q_j} \end{pmatrix}$$

$$Y \mid X = x_i : \begin{pmatrix} y_1 & y_2 & \dots & y_m \\ \frac{\pi_{i1}}{p_i} & \frac{\pi_{i2}}{p_i} & \dots & \frac{\pi_{im}}{p_i} \end{pmatrix}$$

⑧ Obs : !!!

Media unei variabile a. cond. (nu) trebuie confundată cu media condiționată
↓ număr ↓ var. aleatoare

X \ Y	-2	-1	0	1	p_i
-1	1/80	2/80	3/80	14/80	1/4
0	2/80	3/80	14/80	1/80	1/4
1	3/80	14/80	1/80	2/80	1/4
2	14/80	1/80	2/80	3/80	1/4
q_j	1/4	1/4	1/4	1/4	1

$$X : \begin{pmatrix} -1 & 0 & 1 & 2 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}$$

$$Y : \begin{pmatrix} -2 & -1 & 0 & 1 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}$$

c) $X \mid Y=0 : \begin{pmatrix} -1 & 0 & 1 & 2 \\ \frac{3}{20} & \frac{14}{20} & \frac{1}{20} & \frac{2}{20} \end{pmatrix}$

$Y \mid X=2 : \begin{pmatrix} -2 & -1 & 0 & 1 \\ \frac{14}{20} & \frac{1}{20} & \frac{2}{20} & \frac{3}{20} \end{pmatrix}$

$\mathbb{E}[X \mid Y=0] = \frac{-3}{20} + \frac{1}{20} + \frac{4}{20} = \frac{1}{10}$

$\mathbb{E}[Y \mid X=2] = \frac{-28}{20} - \frac{1}{20} + \frac{3}{20} = -\frac{13}{10}$

Proprietățile covarianței

1) $\mathrm{cov}(X, X) = \mathrm{Var}(X)$
2) $\mathrm{cov}(X, Y) = \mathrm{cov}(Y, X)$
3) $\mathrm{cov}(X, c) = 0 \quad \forall c \in \mathbb{R}$
4) $\mathrm{cov}(aX + b \cdot Y, \overbrace{cX + d \cdot Y}^{Z}) = \mathrm{cov}(aX, Z) + \mathrm{cov}(bY, Z)$

$$\begin{aligned} &= a\,\mathrm{cov}(X, Z) + b\,\mathrm{cov}(Y, Z) \\ &= a(\mathrm{cov}(X, cX) + \mathrm{cov}(X, dY)) + b(\mathrm{cov}(Y, cX) + \mathrm{cov}(Y, dY)) \\ &= a(c\,\overbrace{\mathrm{cov}(X, X)}^{\mathrm{Var}(X)} + d\,\underline{\mathrm{cov}(X, Y)}) + b(c\,\underline{\mathrm{cov}(Y, X)} + d\,\overbrace{\mathrm{cov}(Y, Y)}^{\mathrm{Var}(Y)}) \\ &= a \cdot c \cdot \mathrm{Var}(X) + (a \cdot d + b c)\,\mathrm{cov}(X, Y) + b \cdot d\,\mathrm{Var}(Y) \end{aligned}$$

⑧ b') Folosind coef de corelație calculat la punctul anterior $\overset{\rho(X,Y)=k}{\vee}$, calculați $\rho(\overbrace{3X - Y}^{A}, \overbrace{4X + 3}^{B})$

$$\rho(A, B) = \frac{\mathrm{cov}(A, B)}{\sqrt{\mathrm{Var}(A) \cdot \mathrm{Var}(B)}}$$

$$\begin{aligned} \mathrm{cov}(A, B) &= \mathrm{cov}(A, 4X + 3) \\ &= \mathrm{cov}(A, 4X) + \overbrace{\mathrm{cov}(A, 3)}^{0} \\ &= 4\,\mathrm{cov}(A, X) \\ &= 4\,\mathrm{cov}(3X - Y, X) \\ &= 4(\mathrm{cov}(3X, X) + \mathrm{cov}(Y, X)) \\ &= 4(3\,\overbrace{\mathrm{cov}(X, X)}^{\mathrm{Var}(X)} - \mathrm{cov}(Y, X)) \\ &= 12\,\mathrm{Var}(X) - 4\,\mathrm{cov}(X, Y) \end{aligned}$$

$$\left[\rho(X, Y) = k \Leftrightarrow \frac{\mathrm{cov}(X, Y)}{\sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)}} = k \Rightarrow \mathrm{cov}(X, Y) = k \cdot \sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)} \right]$$

$$= 12\,\mathrm{Var}(X) - 4k\sqrt{\mathrm{Var}(X) \cdot \mathrm{Var}(Y)}$$

$\mathrm{Var}(A) = \mathrm{Var}(3X - Y) =$

Obs Atunci când X și Y nu știm că sunt independente :

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{cov}(X, Y)$$

$$\begin{aligned} &= 9\,\mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{cov}(3X, Y) \\ &= 9\,\mathrm{Var}(X) + \mathrm{Var}(Y) - 6\,\mathrm{cov}(X, Y) \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(B) &= \mathrm{Var}(4X + 3) \\ &= 16\,\mathrm{Var}(X) \end{aligned}$$

T: de continuat

b'') Calculați $\mathbb{P}(X \ge 1, Y < 0) = \frac{3}{80} + \frac{14}{80} + \frac{14}{80} + \frac{1}{80} = \frac{32}{80}$